

CyClaw-Net-Viewer

User Guide

*A TCPView-class live TCP/UDP endpoint viewer for macOS,
built for CyClaw telemetry-kill verification.*

Version 0.1.0 · macOS 12+ (Monterey) · Apple Silicon and Intel · MIT License

This guide describes how to install, read, and use CyClaw-Net-Viewer to watch local TCP and UDP sockets, including off-box remotes that matter when proving a telemetry kill switch. It is not affiliated with Microsoft, Sysinternals, or TCPView. Windows TCPView remains the inspiration for the table and the color events; Darwin cannot copy every Windows kernel trick, and those limits are called out where they matter.

1. What this app is

CyClaw-Net-Viewer is a native macOS desktop tool that lists every TCP and UDP endpoint it can see on this Mac, the process that owns it, and whether the peer is on-box or off-box. It refreshes about once a second (adjustable) and paints rows when sockets appear, change state, disappear, or talk to a remote address.

Think of it as a more readable, live subset of what `netstat` and `lsof -i` already know, with Sysinternals-style highlighting. The crate and binary are still named `netboard`; the window, bundle, and this guide use **CyClaw-Net-Viewer**.

It is for

- Watching which apps hold ESTABLISHED TCP sessions to the internet.
- Confirming CyClaw (or any local process) is not opening unexpected off-box sockets.
- Catching short-lived connects: new rows flash green, closed rows linger red.
- Exporting a CSV snapshot for notes or a ticket.

It is not

- A packet sniffer. It does not show ICMP, ping, or payload.
- A firewall. It cannot block a flow.
- A perfect clone of Windows TCPView's "Close Connection." Darwin has no public TCB-delete for another process's socket.
- A cloud service. Nothing is uploaded. Reverse DNS is a local lookup.

2. Install and first launch

Requirements: macOS 12 (Monterey) or newer, Intel or Apple Silicon. The shipped `.app` is a universal binary (arm64 + x86_64) with deployment target 12.0. Do not enable App Sandbox or Hardened Runtime; those hide other processes' sockets.

Double-click the app

The packaged app is `dist/CyClaw-Net-Viewer.app` in the project, or a DMG from `./scripts/make-app.sh --dmg`. The binary is ad-hoc signed, not notarized. Gatekeeper will block the first double-click.

- Finder: Right-click `CyClaw-Net-Viewer.app` → Open → Open.
- After that, ordinary double-click works for this user account.
- If macOS still refuses, System Settings → Privacy & Security → Open Anyway.

Build from source

Rust 1.85.0 is pinned (`rust-toolchain.toml`). Homebrew's newer `rustc` is not the pin. Put `rustup` on `PATH`, then:

```
export PATH="$HOME/.cargo/bin:$PATH"
cd Mac-NetViewer-EZview
cargo run
cargo run -- --cli -n -a
./scripts/make-app.sh
```

Viewing never requires root. SIP-protected processes may omit sockets unless you later choose to run as root; the table is then just more complete, not a different product.

3. Window tour

The window title is **CyClaw-Net-Viewer**. A menu bar (File, Options, Help), a one-line toolbar, a sortable table, and a status bar with a color legend fill the rest.

Toolbar

Control	What it does
Pause	Stop the one-second snapshot loop. Space also toggles pause unless the filter box is focused.
Refresh	Take a snapshot now. Key R.
Resolve names	Default on. Shows hostname (IPv4):port when PTR/A exist. Key N.
Color remotes	Default on. Stable off-box TCP/UDP stays orange.
Off-box only	Hide listeners, loopback, and *:*. Use this for telemetry-kill watching.
Rate	0.5s, 1s (default), 2s, or 5s.
Filter	Case-insensitive substring on process, PID, proto, direction, addresses, state, path.

Columns

Column	Meaning
Process	Short name from proc_name. “?” if the PID could not be named.
PID	Owning process id. 0 means the kernel did not attach a PID.
Proto	TCP4, TCP6, UDP4, or UDP6.
Dir	Listen, In (accepted on a local listener), or Out (including SYN_SENT).
Local	Local address and port. Unspecified shows as *:port.
Remote	Peer address and port. Off-box remotes are the telemetry-kill signal.
State	TCP FSM name (ESTABLISHED, LISTEN, TIME_WAIT, ...). Empty for UDP.
Path	Full executable path from proc_pidpath, when available.

Click a header to sort; click again to reverse. Click a row to select. Right-click for Copy line, Copy remote, Terminate process, and Reveal in Finder. The status bar shows endpoint count, how many are off-box, and the color legend.

4. Color scheme

Event colors follow Sysinternals TCPView. CyClaw-Net-Viewer adds a persistent orange for any TCP/UDP peer that is not unspecified and not loopback, so a phone-home that is already ESTABLISHED does not look like a blank row. Event colors always win over orange.

Swatch	Name	When you see it
	New out	New outgoing endpoint since the last tick (including SYN_SENT).
	New in	New incoming endpoint or new LISTEN.
	Changed	Same 4-tuple, TCP state changed (for example ESTABLISHED → CLOSE_WAIT).
	Gone	Endpoint vanished. Shown for two refresh ticks (~2s at 1s rate), then dropped.
	Off-box	Stable TCP/UDP to a remote that is not this Mac. Stays lit while the socket exists.

Priority if more than one would apply: gone > new > changed > off-box. A row cannot be both new and gone. Loopback and *:port stay uncolored when stable so the orange set is actually egress.

ICMP and ping never appear. They are not TCP or UDP sockets. To exercise colors, use `curl, nc host 443`, or start the process under test.

5. Addresses and DNS

With Resolve names on, each cell prefers a human form. Reverse DNS (PTR) runs on a background pool. If PTR returns a hostname, a forward A lookup tries to attach an IPv4. IPv6 sockets stay in the table (they are real); the cell can still show an IPv4 next to the name when DNS has one.

What DNS knows	Cell looks like
Hostname and IPv4 (socket is v4, or A record from the name)	mail.google.com (142.250.105.101):993
Hostname, IPv6 only	mail.google.com ([2607:f8b0:...]):993
IPv4-mapped IPv6, no name	8.8.8.8:443
No PTR	Numeric IPv4 or compressed IPv6 with port
Unspecified	*:443

Dual-stack apps often show two rows (TCP4 and TCP6) to the same service. That is correct, not a duplicate bug. Uncheck Resolve names (or press N) for pure numerics. Failed lookups are not retried until the app restarts.

6. CyClaw telemetry-kill recipe

CyClaw's telemetry-kill switch is supposed to stop vendor SDKs from phoning home. This app is the eyeball check on the socket table. Do not turn telemetry on to test the kill switch. Watch whether CyClaw's process grows off-box rows.

Setup

- Launch CyClaw-Net-Viewer. Turn on Off-box only and Color remotes.
- Set rate to 0.5s or 1s. Clear the filter, or set it to the CyClaw process name or PID.
- Note the orange baseline: browsers, Mail, and the system will already have remotes. That is normal.

Run the subject

- Start CyClaw (or the binary under test) with telemetry-kill as you would in production.
- Watch for new green Out rows whose Path is the CyClaw binary, then orange while they live.
- Filter on the CyClaw PID once you know it (Activity Monitor or the Path column).

How to read the result

What you see	What it means
No new orange/green rows whose Path is CyClaw	Socket-level pass for this run. Pair with the static otel-hardening checker; sockets are not the whole story (DNS-over-HTTPS inside another process, delayed connects).
Green then orange to a vendor host from CyClaw	Kill switch did not hold for that path. Copy the line (right-click) and keep the remote hostname/IPv4.
Red after you quit CyClaw	Those sockets closed. Expected. Red lingers two ticks.
Ping to the vendor lights nothing	Expected. Ping is ICMP. Use curl or the real process.

Always keep the static CyClaw checker in the loop. In a CyClaw checkout run:

```
python .claude/skills/otel-hardening/check_otel.py
```

This GUI only sees sockets that exist on this Mac at refresh time. Sockets are not the whole story: delayed connects and traffic inside another process will not show as CyClaw.

7. Keyboard and menus

Key / menu	Action
Space	Pause / resume (ignored while the filter box is focused).
R	Refresh now.
N	Toggle Resolve names.
Delete or Backspace	Terminate the selected process, after confirm.
Click header	Sort; click again to reverse.
File → Save CSV	Write cyclaw-net-viewer-YYYYMMDD-HHMMSS.csv in the current working directory (often your home if you launched the .app from Finder).
File → Close Connection	Enabled only for a selected ESTABLISHED TCP row. Same as terminate process.
File → Quit	Exit.
Options	Resolve names, Show listeners, Show UDP, Color remotes, Off-box only, refresh rate.
Help → About	Version, color summary, Darwin TCB note.

8. Close Connection on Darwin

Windows TCPView can abort another process's ESTABLISHED TCP by deleting the kernel TCB (SetTcpEntry). macOS has no public equivalent that is safe and unsandboxed. CyClaw-Net-Viewer therefore asks you to confirm SIGTERM of the owning PID. That ends the whole process, not just one socket.

- The dialog states this clearly. Cancel if you only meant to drop one connection.
- The app refuses PID 0 and refuses to signal itself.
- There is no RST injector and no sudo helper in this version.

9. Command-line mode (Tcpvcon-style)

The same binary prints a snapshot without the GUI. Default is ESTABLISHED TCP only, matching Tcpvcon. Reverse DNS is on unless you pass `-n`.

```
netboard --cli [-a] [-c] [-n] [process|pid]

-a    all TCP and UDP endpoints
-c    CSV (Process,PID,Proto,Dir,Local,Remote,State,Path)
-n    numeric addresses, no DNS
filter optional PID (digits) or process-name substring
```

Examples: `netboard --cli -n -a` for a fast full dump; `netboard --cli -c python` for CSV of processes whose name contains “python”. GUI is the default when `--cli` is omitted.

10. Limits and known issues

Limit	Detail
SIP / TCC	Protected processes often omit FDs without root. The app still runs unprivileged.
UDP remotes	The socket enumerator does not expose connected UDP peers. Those rows look like <code>*:0 / Listen</code> .
No byte counters	Darwin <code>socket_info</code> does not give cheap cumulative bytes without packet tap.
First tick	Walking every PID can take a beat; then it settles at the chosen rate.
Unsigned app	Ad-hoc codesign only. Notarization is out of scope.
Mutex / panic	If a snapshot panics, the last good rows stay on screen and the status bar reports it.

11. Optional later work (not in 0.1.0)

The app is enough for live watching and a kill-switch eyeball check. If you want more later, these are the useful ones — not a promise they will ship:

- Remember Off-box only, rate, and window size across launches.
- Filter by executable path, not only short process name.
- Pair dual-stack TCP4/TCP6 to the same peer on one row (without dropping IPv6).
- Connected UDP remotes if a future enumerator exposes them.
- Notarized Developer ID build for painless Gatekeeper.
- Optional root RST closer (still not a real TCB delete).

Do not add ICMP to the table, App Sandbox, or a webview rewrite. Those fight the point of a local, unsandboxed socket list.

12. Credits

Inspiration: Sysinternals TCPView by Mark Russinovich. This tool is an independent macOS implementation under the MIT License and is not affiliated with Microsoft. Socket listing uses Darwin libproc (the same family of APIs as lsof). Source lives in this repository. Crate name: `netboard`.